



Alejandro Arrua

Systems Analyst | Senior Full Stack Developer | Tech Lead

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🌐 <https://portfolio-next-js-git-master-arrua-alejandros-projects.vercel.app/es>

Summary

Senior Full Stack Developer and Tech Lead with more than 8 years of experience designing scalable software architectures (Microservices, MFE, DDD) and leading multidisciplinary technical teams. Polymath developer with high adaptability, mastering ecosystems ranging from Node.js, PHP/Laravel, and React, to C# (.NET), Java (Spring Boot), Python (IoT/IA), and C++ (QT). Proven track record in AI integration, enterprise database migration, DevOps (AWS, Kubernetes), and compliance with security regulations (ISO/IEC 27001). Oriented towards code quality (SOLID, Clean Architecture) and strategic business impact.

Education

UADER - FCyT	Systems Licentiate
Paraná, Entre Ríos, ARG • March 2016 – In progress	
UADER - FCyT	Systems Analyst
Paraná, Entre Ríos, ARG • March 2016 – December 2022	
Henry Bootcamp	Full Stack Web Developer
Remote • March 2022 – July 2022	

Experience

Tecnosoftware	Remoto
Senior Systems Analyst / Full Stack Developer	October 2024 - Present
• Microservices Architecture: Led the development of a Document Management System (DMS) and its BackOffice for a global automotive brand, implementing IaaS, BaaS, and MFE architectures. • Modeling & Database: Data modeling with Extensible DER (StarUML). Successful migration of instances and custom databases from SQL Server to PostgreSQL, developing solutions in PL/SQL (views, stored procedures, functions, and transactions). • Stack & Tools: Utilization of WebHooks, WebSockets, Redis, MongoDB (Compass), Pusher, AWS Lambda with Node.js, Cloudflare, API Gateway, Eventing, Nginx, AWS X-Ray, and Sentry. Frontend with Metronic V8 and Select2. • Modernization & Quality: Migration of microservices from Slim 3/PHP 7 to Slim 4/PHP 8 applying the ADR pattern. Development of version-controlled libraries, applying SOLID principles and design patterns. • Management & Compliance: Use of Bitbucket and Jira under Scrum methodology. Implementation of documentation throughout the product lifecycle, ensuring adherence to security standards ISO/IEC 27001 and ISO/IEC 27034.	
Invisible Geeks	Remote
Software Engineer (AI & SaaS Focus)	August 2024 - Present
• AI Integration: Development of SaaS MVP solutions for renowned clients (e.g., Johnson & Johnson), integrating artificial intelligence models (OpenAI in all its versions, Gemini, Mistral AI) and creation of AI-based chats. • Scraping & Optimization: Architecture of a data extraction solution (scraping) for the tourism sector (travel, flights, leisure), applying advanced caching strategies, middlewares, and performance testing to identify refactoring opportunities.	

- **Full Stack & Mobile Development:** Construction of applications with Laravel Inertia + React, including continuous deployment (CI/CD), unit testing, code review, and development of mobile apps with access to native components (camera, sensors) and binary file handling.
- **Requirements Engineering:** Active participation in meetings with international stakeholders to understand business needs and propose strategic product improvements, complemented by DevOps tasks.

NewBitCrew	Remote
Full Stack Developer / Tech Lead (Project B1U)	December 2022 - April 2024
Analysis and development in: -PHP with Laravel (6-10), Laravel Mix (v6-v10), Symfony. phpMyAdmin/Valentina Studios as database manager. Application of OOP, design patterns, SDKs for other platforms, secure API-REST modules, utilization of the search engine Meilisearch, Elasticsearch, configuration of Events, Policies, Repositories, Models, Services, Helpers, Middlewares, Providers, Cache Methods, Configurations, application of Eloquent and efficient SQL statements, blade and twig templates. -JavaScript/Typescript, with Vanilla, Next.js, Vue.js, Angular, Nest, React with Redux, Redux-Toolkit, Formik, Universal-cookie Node.js with Express JS, JWT, Sequelize-Cli. Database manager Adminer. CMS development. Paywall. -C# with <u>ASP.NET</u> , .NET CORE 6/8, Kafka Confluent, MailGun, Mercado Pago SDK, Cache methods, JWT, Razor, WPF, EF, among other technologies. -Java with Spring Boot, Mercado Pago SDK, RabbitMQ, JWT, Cache methods. -Application of Hexagonal Architecture, DDD and IoT applications. -Development in Python with Selenium, TensorFlow, SQLAlchemy, FastAPI. -Manual, automated (Selenium) and unit testing (PHPUnit, Jest, etc). -Scrum using Trello and Jira. Daily meetings on Discord and Slack. -Docker containers. Linux operating system, use of command line.	

- Projects I participated in:
- Perfil: I developed a news CMS.
- B1U: I developed a virtual classroom platform performing as Tech Leader (.Net + React TS).
- Beathey: I implemented improvements in a music platform.
- Opinaia: I performed manual tests on endpoints using Postman and CURL commands, and monitored the browser console.
- Coin: I applied improvements in a payment platform. Improvements in IoT projects with Python and TypeScript.

General Directorate of Statistics and Censuses	Paraná, Entre Ríos, ARG
Analyst Programme	August 2022 - December 2022
• Backend development in PHP (Laravel) and JS (Node.js); Frontend in JS (React, Redux Toolkit, React Native).	
• Administration, maintenance, migration, and backup of MS SQL Server databases (2004-2019) using VMWare, plus technical support.	

Argentine Army	Paraná, Entre Ríos, ARG
IT Technician & Software Developer	September 2014 - July 2022
• Design and maintenance of LAN networks.	
• Design and development of internal software to digitize manual processes using Laravel Mix + Vanilla JS/React, and .NET applications (online via Razor/ <u>ASP.NET</u> and offline via WPF).	
• Maintenance and evolution of existing projects in Java with Spring Boot.	

Certifications

DevOps TOTAL: Docker, Kubernetes, Jenkins, AWS, Git & Más	2025
Udemy	
Master en PHP, SQL, POO, MVC, Laravel, Symfony, WordPress+	2025
Udemy	
English C1 Advanced	2025
EF SET Certificate	
(65/100 - Reading 68, Listening 62)	

Backend & Languages:

Node.js (Express, NestJS), PHP (Laravel 6-10, Symfony, CodeIgniter, Slim 3/4, PHP 7/8), C# (.NET Core 6/8, ASP.NET, Razor, WPF, Entity Framework), Java (Spring Boot 8+), Python (FastAPI, Selenium, TensorFlow, SQLAlchemy, IoT), C++ (QT).

Frontend & Mobile:

React.js (Redux, Redux Toolkit), Next.js, Vue.js (2 & 3), Angular, TypeScript, JavaScript (Vanilla, Funcional, Componentizado), HTML5, CSS3, Bootstrap, Tailwind CSS, Laravel Mix, Livewire, Jetstream, Nova, Blade, Twig, Webpack, Vite, React Native.

Databases & Search:

PostgreSQL, MS SQL Server (2004-2019), MySQL, MongoDB, Redis, Elasticsearch, Meilisearch, Firebase (Auth), Sequelize, Mongoose. Cloud, DevOps & Infraestructura: AWS (Lambda, EC2, S3, X-Ray), Azure, Google Cloud, Docker, Kubernetes, Jenkins, Terraform, Ansible, Nginx, Linux (CLI), Windows Server, CI/CD (GitHub Actions), Cloudflare, XAMPP, WAMP.

Architecture & Patterns:

Microservicios (IaaS, BaaS, MFE), Domain-Driven Design (DDD), Arquitectura Hexagonal, Patrón ADR, SOLID, POO, ACID, Inyección e Inversión de Dependencias, REST API, WebSockets, WebHooks, Clean Code, Refactoring.

Tools, Testing & Security:

Git (GitHub, GitLab, Bitbucket), Jira, Trello, Postman, Nodemon, Morgan, DBee, Adminer, phpMyAdmin, Valentina Studio, MongoDB Compass, StarUML (DER Extensible). Testing: Jest, PHPUnit, Selenium. Seguridad OWASP: Dependency-Check, ZAP, CSRFGuard, AntiSamy, ESAPI, JWT, Bcrypt, Universal Cookies.

Messaging & External Services:

Kafka (Confluent), RabbitMQ, Socket.IO, Pusher, API Gateway, Eventing, Mailhog, Nodemailer, MailGun, Mercado Pago SDK.